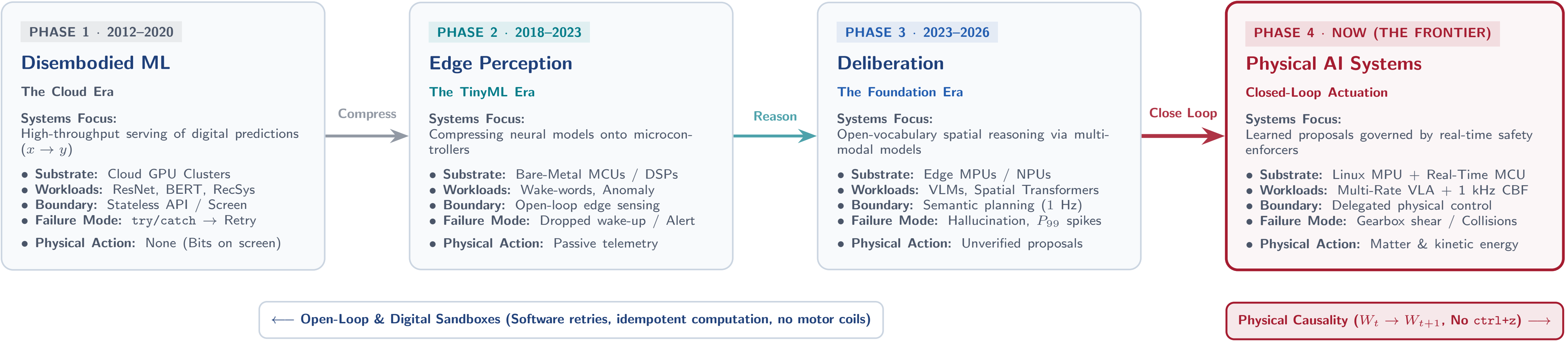
PHASE 1 · 2012–2020
Disembodied ML
The Cloud Era
Systems Focus:
High-throughput serving of digital predictions ($x \to y$)
• Substrate: Cloud GPU Clusters
• Workloads: ResNet, BERT, RecSys
• Boundary: Stateless API / Screen
• Failure Mode: try/catch → Retry
• Physical Action: None (Bits on screen)
Compress
PHASE 2 · 2018–2023
Edge Perception
The TinyML Era
Systems Focus:
Compressing neural models onto microcontrollers
• Substrate: Bare-Metal MCUs / DSPs
• Workloads: Wake-words, Anomaly
• Boundary: Open-loop edge sensing
• Failure Mode: Dropped wake-up / Alert
• Physical Action: Passive telemetry
Reason
PHASE 3 · 2023–2026
Deliberation
The Foundation Era
Systems Focus:
Open-vocabulary spatial reasoning via multimodal models
• Substrate: Edge MPUs / NPUs
• Workloads: VLMs, Spatial Transformers
• Boundary: Semantic planning (1 Hz)
• Failure Mode: Hallucination, P_{99} spikes
• Physical Action: Unverified proposals
Close Loop
PHASE 4 · NOW (THE FRONTIER)
Physical AI Systems
Closed-Loop Actuation
Systems Focus:
Learned proposals governed by real-time safety enforcers
• Substrate: Linux MPU + Real-Time MCU
• Workloads: Multi-Rate VLA + 1 kHz CBF
• Boundary: Delegated physical control
• Failure Mode: Gearbox shear / Collisions
• Physical Action: Matter & kinetic energy
⟵ Open-Loop & Digital Sandboxes (Software retries, idempotent computation, no motor coils)
Physical Causality ($W_t \to W_{t+1}$, No ctrl+z) ⟶